TSM 099 Quadratic and Rational Equations Test - Practice 2

Version 202610122020.26 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Solve by factoring.

$$x^2 + 3x - 10 = 0$$

2. Solve by taking square roots; do not expand the square.

$$2(x)^2 = 18$$

3. Solve. List excluded values and check in the original equation.

$$\frac{6}{2x-4} + \frac{10}{x^2-7x+10} = \frac{4}{x-5}$$

4. Solve with the quadratic formula. Simplify radicals.

$$2x^2 + 2x - 3 = 0$$

5. Solve. List excluded values and check candidates in the original equation.

$$\frac{6}{x^2-9} + \frac{x-4}{x-3} = \frac{7}{x+3}$$

Student answer sheet

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Question 1

Solutions

20 pts

Question 2

Solutions

20 pts

Question 3

Excluded values

5 pts

Solution/check

15 pts

Question 4

Solutions

20 pts

Question 5

Excluded values

4 pts

Candidates/check

6 pts

Solution

10 pts

Answer key

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Question 1

Solutions	$x = -5,\ 2$	20 pts

Question 2

Solutions	$x = -3,\ 3$	20 pts

Question 3

Excluded values	$\{2, 5\}$	5 pts
Solution/check	$x = 3$	15 pts

Question 4

Solutions	$x = -\frac{1}{2} + \frac{\sqrt{7}}{2},\ \ -\frac{\sqrt{7}}{2} - \frac{1}{2}$	20 pts

Question 5

Excluded values	$\{-3, 3\}$	4 pts
Candidates/check	$x = 3$ (rejected), $x = 5$	6 pts
Solution	$x = 5$	10 pts